Kasa Integration 2.5.3 Documentation

This version of the Kasa integration incorporates all Kasa and Tapo branded plugs, switches, bulbs, and light strips. This document contains several sections:

a. Tapo Integration App description
b. Device pages description
c. Release notes for this version from previous.

Tapo Integration App Description

Startup Page

a. **Quick Installation Instructions:** Displays when **Display Start Page Help** is selected at the bottom of the page.
b. **Set Up Application and Credentials:** Required for Tapo-branded devices and for Kasa devices that have transitioned to the Tapo API (smart) or Tapo protocol (KLAP). When credentials are entered:
    - Device scans include devices that use the Tapo protocol.
    - Quick Scan, including error-recovery scanning, also includes Tapo protocol devices.
c. **Remove Devices:** Opens the Remove Devices page, where installed devices can be selected for removal.
d. **Debug Logging:** Enables detailed debug logging for 30 minutes.
e. **Information Logging:** Enables informational log messages. Keeping this option enabled is recommended.
f. **Display Start Page Help:** Shows or hides the quick installation instructions.

Set Up Application and Credentials Page

a. **Quick Instructions:** Displays when **Display Setup Help** is selected at the bottom of the page.
b. **Account and Installation Settings:** Enter the account email address and password, and select whether to install Matter-capable devices.
c. **LAN Segments and Kasa IoT-Only Ports:** These settings apply only to legacy installed devices and do not work with newer devices.
    - **Important:** Some legacy devices may not transition successfully when they are not on the default LAN segment.
d. **Display Setup Help:** Shows or hides the setup instructions.

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Add Devices Page

After a device scan, use this page to select and install devices:

1. Choose a device from the drop-down list.
2. If an expected device is not listed, select **Rescan for Additional Devices**.
3. After making your selections, select **Next**.

Important Notes

- **Legacy devices:** Some legacy Kasa devices enter deep sleep. If an expected device is missing, operate it in the Kasa/Tapo mobile app, and then select **Rescan for Additional Devices**.
- **DHCP reservations:** Operation depends on DHCP reservations configured on your local router. Verify that each device has a reservation.

Device Pages Description

Each device uses the capabilities defined by its driver. For most capabilities, available actions follow the Hubitat Capabilities documentation. The device-page customizations are described below.

Commands

- **Configure:** Checks for the device at its configured IP address and expected port. If the device is not found, the driver requests an application-wide device scan, which updates the base URL and protocol when either has changed.
- **Refresh:** Retrieves current device information. If energy monitoring is enabled, it also retrieves the current power reading.
- **Set Lights With Transition:** Provides method for rule machine to set on/off and color with a instance-specific transition time. Command format is comma delimited Key:Value pair without brackets. Examples:
  - colorTrans: “rate: 1000, level: 55, hue: 22, saturation: 55”
  - colorTempTrans: “rate: 1000, level: 55, color_temp: 4000”
  - levelTrans: “rate: 1000, level, 55”

Current States

- **Comms Error:** Indicates that automatic attempts to correct a communication problem were unsuccessful. The state remains set until communication succeeds.

State Variables

- **Cmd Data:** Stores the most recently sent command for error recovery.
- **Error Count:** Tracks the number of errors.
- **Event Type:** Determines the **eventType** value used by the **sendEvent** command.
- **Retry:** Set by each command-calling method to indicate whether that method will handle communication errors.
- **UpdProcessed:** Used by **updateDeviceData** to prevent unnecessary device-data updates. It is set to five minutes after the most recent update.

Preferences Page

Most preferences are self-explanatory. The following troubleshooting option requires special care:

- **If the device does not work:** Use this option to switch between the KLAP and XOR protocols. If the alternate protocol does not resolve the problem, switch back to the original protocol.

  Caution: If the device does not work with the alternate protocol and cannot be restored to the original protocol, recovery may require removing and reinstalling the device.

Release Notes

This release introduces a major architectural update. The integration now supports two APIs—IoT (legacy) and SMART (new)—and three protocols: XOR (legacy), KLAP (new), and AES (new but deprecated).

Transition Notes

Scheduled-action messages: Early log entries may contain a missing-method message similar to the following because a previously scheduled action ran after its associated method was changed:

> *org.codehaus.groovy.runtime.metaclass.MissingMethodExceptionNoStack: No signature of method: user_driver_davegut_Kasa_Plug_Switch_2716.poll() is applicable for argument types: () values: []. Possible solutions: off(), on(), iotPoll(), wait(), run(), run() (method poll)*

Device warnings: Each device displays two warning messages prompting the user to verify device operation and review the resolved issues below.

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Resolved Issues

• Added support for repackaged Kasa devices that use the Tapo API and protocol.

• Reduced the risk of device incompatibility if TP-Link replaces the legacy XOR protocol with the Tapo KLAP protocol.

• Added compatibility for Hubitat Color Animation.

Open Issues

• On legacy hardware version 1 HS300 devices, UDP truncation requires users to limit the names of the six outlets.

• The deprecated Tapo AES protocol has not been tested.

• A firmware issue on the Tapo P100 plug causes LED updates to fail.

Newly Supported Devices

• Kasa plugs, switches, bulbs, and light strips that use the Tapo API and/or KLAP protocol.

• Tapo devices that connect directly to the network rather than through a separate Kasa or Tapo hub.

User Functionality Changes

• Replaced the **setPollInterval** command with a polling-interval preference.

• Consolidated and updated preferences. Removed the Manual IP, Manual Port, and Reboot Device preferences.

• Added **setLightWithTransition**, allowing Rule Machine to send on/off and color commands with an instance-specific transition time.

• The transition time is applied to the specific command instance.

• Added automatic migration from the legacy API and protocol to the new system when legacy communication fails.

• Added a preference that allows users to switch manually to the new protocol and switch back if the change is unsuccessful.

Application Changes

• Added the **Set Up Application and Credentials** pages. Setup is required for enhanced features but otherwise remains optional.

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• Added optional help text to the startup and application-setup pages.

• Added discovery of devices that use the Tapo API and/or protocol when account credentials have been entered. IP-address updates are not part of this transition.